jMeter PERFORMANCE TEST

For TA.TS - 2023, 2024

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Objective

TA.TS approached AMVARA for seeing 503 errors and performance issues on Dashboard loading of IBIZA Dashboard running on IBM Cognos 11.1.7 FP10

A jMeter test plan to imitate the user story has been developed and executed. Analysis of the same was provided.

This documentation describes how to install, execute, and analyse the jMeter load tests that have been prepared.

This documentation and related was done by: Ralf Roeber and Anand Kushwaha.

jMeter setup in Linux

The following assumes that jMeter is being installed in a Linux environment or a Virtual Box running Linux (e.g. UbuntuServer from OSBOXES.org).

For execution on Windows the installation is slightly different and the run.sh command must be adapted to powershell.

Prerequisites

1. Java installation:

- Download and install the latest Java Development Kit (JDK) from the official Oracle website or adopt OpenJDK.
- Use any Java version between 11.x.x to 17.x.x.
- Set the JAVA_HOME environment variable to point to the JDK installation directory, and add the bin directory to the system's PATH.

2. Check java installation:

- Open a terminal.
- Run the following command to check if Java is installed:
- java -version

3. Determine java installation directory:

- If Java is installed, note the installation directory. Common locations include `/usr/lib/jvm/` or `/usr/local/java/`.

4. Set java home:

- Open the terminal and edit the shell profile configuration file. This could be `~/.bashrc`, `~/.bash_profile`, or `~/.zshrc` depending on your shell.

```
nano ~/.bashrc
```

- Add the following line at the end of the file, replacing /path/to/your/java with your Java installation directory:

```
export JAVA_HOME=/path/to/your/java
```

5. Update path variable:

- Still in the same file, add the following line to include Java's bin directory in the PATH:

```
export PATH=$JAVA_HOME/bin:$PATH
```

6. Save and Exit:

- Save the changes by pressing Ctrl + X, then Y to confirm, and enter to exit.

7. Load the updated configuration:

- To apply the changes, either close and reopen the terminal or run:

```
source ~/.bashrc
```

8. Verify java home:

- Verify that JAVA_HOME is set correctly:

```
echo $JAVA_HOME
```

9. Verify Java Version:
    - Check if Java is now using the correct version:
    - java -version

jMeter installation

1. Download jMeter
    - Visit the official Apache jMeter download page: https://jmeter.apache.org/download_jmeter.cgi
    - Download the latest version of the binary zip file (e.g., apache-jmeter-x.x.zip).

2. Extract jMeter
    - Extract the downloaded zip file to a directory of your choice.

3. Run jMeter
    - Navigate to the jMeter bin directory (e.g., apache-jmeter-x.x/bin).
    - Run jmeter.bat (for Windows) or jmeter.sh (for Unix-based systems) to start jMeter.

4. jMeter interface
    - The jMeter GUI should now open. It might take a moment to initialize.

5. Optional steps

    Environment variables (Optional):
    - Add the jMeter bin directory to the system's PATH environment variable to run jMeter from any command prompt.

    Configure java heap size (Optional):
    - If needed, you can adjust jMeter's memory settings by modifying the jmeter.bat or jmeter.sh file in the bin directory. Look for lines containing set HEAP (for Windows) or HEAP (for Unix-based systems) and adjust the memory settings as necessary.

6. jMeter installation verification
    - To ensure that jMeter is installed correctly, you can open a command prompt (or terminal) and run the following command.

    ```
    jmeter -v
    ```

    - This command should display the version of jMeter, confirming a successful installation.

Prepare and Execute jMeter Tests

Place run script inside jMeter home directory

Assume **JMETER_HOME = /path/to/your/apache-jmeter-x.x** directory.

- Place `run.sh` file inside JMETER_HOME home directory.

Your directory structure should look like this.

```
debia                                r/apache-jmeter-5.6.2$ ls -al
total 88
drwxr-xr-x  9 debian debian  4096 Jan 11 14:37 .
drwxr-xr-x  3 debian debian  4096 Jan 11 14:36 ..
drwxr-xr-x  5 debian debian  4096 Jul  4  2023 bin
drwxr-xr-x  5 debian debian  4096 Jul 11  2023 docs
drwxr-xr-x  2 debian debian  4096 Apr 22  2023 extras
drwxr-xr-x  4 debian debian 12288 Jun 29  2023 lib
-rw-r--r--  1 debian debian 17515 Jun 29  2023 LICENSE
drwxr-xr-x 68 debian debian  4096 Jun 17  2023 licenses
-rw-r--r--  1 debian debian   172 Jun 29  2023 NOTICE
drwxr-xr-x  6 debian debian  4096 Jul 11  2023 printable_docs
-rw-r--r--  1 debian debian 11046 Jun 29  2023 README.md
-rwxr-xr-x  1 debian debian  5284 Dec  8 22:26 run.sh
drwxr-xr-x  2 debian debian  4096 Dec  9 18:03 test_plans
```

- Create folder with name ***test_plans*** in side JMETER_HOME directory.

  Note: test_plans directory contains jMeter test_files(.jmx) file.
  Keep your all test files inside this directory.

Configuration options

- To perform a load test from the command line interface (CLI), utilize the run.sh file along with specified configuration values.

For example:

```
./run.sh -p="Password123" -t 3 -l 2 -rup 60 -s MyTestServer -e TestEnvironment
```

The run.sh file may include multiple configuration parameters, some of which are mandatory.

Note: Mandatory parameters are highlighted.

-p=<password> --password=<password>	Password for the user configured in the JMX file
-p --password	If used with password will be prompted during execution
-t <number> --threads <number>	Number of threads to start. Each thread represents a user.
-l <number> --loops <number>	Number of loops to run. Each loop will run once a report configured.

`-e <environment>, --environment <environment>`	Environment name, must be same as the file name <SERVER>_<ENV>.jmx
`-s <server> --server <server>`	Server name, must be same as the file name <SERVER>_<ENV>.jmx
`-J<option>=<value>`	Additional options that will be passed to the jMeter.
`-rup <number>, --ramp-up-time <number>`	Set the Ramp Up Time in seconds.

1. Examples to prepare configuration options

To initiate a load test, execute the run.sh file with the specified configuration values below.

#1 --

- Number of Users: 3,
- Ramp up Time: 60 Seconds,
- Loops: 2
- Password: Password123
- JMeter Test File Name: MyTestServer_TestEnvironment.jmx

The command structure should be as follows.

```
./run.sh -p="Password123" -t 3 -l 2 -rup 60 -s MyTestServer -e
TestEnvironment
```

#2 --

- Number of Users: 50,
- Ramp up Time: 180 Seconds,
- Loops: 3
- Password: 123Password
- jMeter Test File Name: DemoServer_EnvTest.jmx

The command structure should be as follows.

```
./run.sh -p="123Password" -t 3 -l 2 -rup 60 -s DemoServer -e
EnvTest
```

How to execute

Prepare your JMX file and place it inside the ***JMETER_HOME/test_plans*** directory, open a Bash terminal and navigate to the ***JMETER_HOME*** directory.

Note: Keep your run.sh file as mentioned in the **Place Script** section.

- Prepare your test plan as mentioned in the Configuration Options > Examples above.
- Run your command.

```
~/Downloads/Apache-Jmeter/apache-jmeter-5.6.2$ ./run.sh -p="E[redacted]" -t 3 -l 2 -rup 120 -s T[redacted]T -e I[redacted]T_V2
:30:22][INFO][run.sh:104] - Running jMeter Script
:30:22][INFO][run.sh:105] - Checking required parameters
:30:22][INFO][run.sh:118] - Making some directories
:30:22][INFO][run.sh:29] - Running the test plan
:30:22][INFO][run.sh:54] - Tailing the logs:
```

How to read results

- After the test execution is complete, a test result directory containing the HTML report will be generated inside ***JMETER_HOME/logs_dashboard***.
- The test result directory name will follow the format.

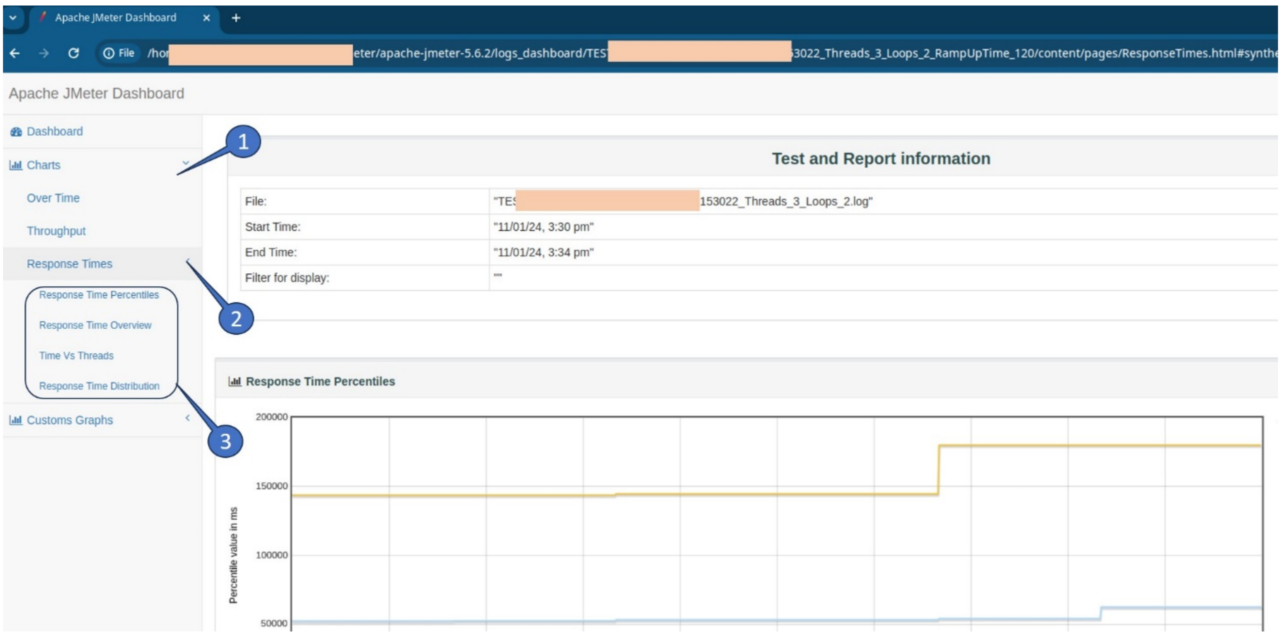
```
{Jmeter-test-file-name}_{Date_YYYYMMDD}_{Time_HHMMSS}
_Threads_{number-of-threads}_Loops_{number_of_loops}
_RampUpTime_{ramp-up-time}
```

- Within the directory, open the index.html file in a web browser.

Analyse the execution report

- The dashboard includes a **Request Summary Graph**, an **Application Performance Index**, and a **Statistics table**. Please refer to the details for more information about the test.
- Additionally, you can view the execution graphs by following these steps

  1. Charts
  2. Response Times
  3. Select required Graph



AMVARA

https://www.amvara.de/

info@amvara.de

+34 717 102 603